

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0

- **Coursework:** Data Structures & Algorithms, Discrete Mathematics

EXPERIENCE

Vylet

May 2026 – Present | vyletdata.com | GitHub

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels, then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Diagnosed a name-collision defect that rejected valid targets sharing a name with an unrelated business; the fix lifted lead-qualification from 79% to 89% with zero change to sourcing volume.
- Grew Vylet from 1 to 3 paying subscription clients within six weeks of launch — spanning workforce-software, landscaping, and geography-first sourcing engagements — generating \$1,500 in monthly recurring revenue.

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

PROJECTS

SignalWeaver | *LangGraph, FastAPI, pgvector, React, TypeScript*

github.com/Verdent06/SignalWeaver

- Orchestrated a 5-node LangGraph pipeline (fetch → classify → embed → score → explain) with per-node timing from the same 90-run batch: fetch 7.3s p50, classify 1.1s p50 (~29ms/article over 40-article batches), embed 1.1s p50, score under 1ms p50, explain 3ms p50 — fetch dominated wall time (~80% of p50 latency).
- Implemented semantic search over stored financial news with pgvector cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Built a React/TypeScript dashboard for composite score breakdowns; analyzed 90 distinct tickers with scores persisted to Postgres for history views.
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL
Frameworks	React, Angular, FastAPI, Flask, RxJS
AI / ML	LangGraph, LangSmith, Silero VAD, ONNX Runtime, Pandas
Databases	pgvector, PostgreSQL
Cloud & Infrastructure	AWS (EC2), Docker, Redis, Celery, Git